

ANKIT SINGH

Full-Stack Platform Engineer | API & System Performance | AWS | Reliability, Observability & Incident Response

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Lead engineer with 6+ years of experience building and scaling customer-facing systems and APIs in fintech and cloud environments. Strong in system design, data modeling, reliability/observability, and performance optimization. Experienced in end-to-end architecture and production operations (monitoring, incident response, cost and performance tuning) and working with product/ops to ship useful features. I enjoy contributing to open source projects in my free time.

Senior Software Engineer — Sequoia

July 2024 - Present

Project: GenAI platform (Dify + RAG)

- Built an internal GenAI agent platform (Dify + RAG) that multiple teams reuse, with shared configs, runbooks, and guardrails, using Python, FastAPI and MongoDB. Focused on the relevance of retrieved context and quality of responses for internal users.
- Refined prompts and retrieval in Dify’s Prompt IDE to make answers more relevant and reduce bad responses.
- Led technical decision-making and architectural tradeoffs, aligned engineering work with business outcomes and system constraints.

Lead Engineer — LotusPay (acquired by Juspay)

February 2020 - June 2024

Project: Developing India's leading NACH mandate and payments system

Public Platform & Developer Experience

- Owned payment dashboards using Ruby on Rails and JavaScript for CRUD payment operations, simplifying workflows for support and ops teams.
- Developed targeted payment tracking pop-ups, improving user clarity and increasing payment completion from 69% to 90% through enhanced visibility into transaction statuses.
- Optimized AJAX requests through HTTP/2, enabled keep-alive, and batched requests, helping reduce page load time from 7s to 1s, delivering a smoother user experience.

Reliability, Observability & Incident Response

- Engineered scalable REST APIs (Rails, Redis, AWS EC2), improving payment success rate from 50% to 90%; implemented consistent error schemas, pagination, and API versioning to reduce integration issues and support escalations.
- Improved API developer experience by standardizing error responses and adding structured logs, cutting customer confusion and lifting CSAT by 25%.
- Fixed critical endpoints with idempotency keys and strict request validation to prevent duplicate processing under retries.
- Introduced retry-safe processing, traceable logging, and audit-friendly safeguards in core payment flows, plus data-driven changes that increased conversions by 30%.
- Improved performance with caching, tuned DB queries/indexes, and blue-green deployments, and set up CloudWatch + SNS + Lambda alerts to cut downtime by 95% and keep incident resolution under 5 minutes.
- Reworked our bank payment data ingestion pipeline by running A/B tests on batch size, indexing, and concurrency settings on AWS RDS (MySQL), the winning setup cut processing time by 15x and made imports far more reliable under load.

Performance & Release Engineering

- Redesigned the core MySQL schema on AWS RDS, tuned heavy queries (New Relic + EXPLAIN), and added Redis + CDN caching, cutting server load by 50% and saving over \$10k annually.
- Implemented blue-green deployments using Route 53 + S3 + CloudFront, enabling zero-downtime rollouts and reducing release-related disruptions by 30% during high-traffic hours.
- Led quarterly reliability and performance efforts and mentored engineers through design and code reviews, improving code quality and release practices.

Skills

- Ruby on Rails, Python, Node.js
 - MySQL, PostgreSQL, MongoDB
 - HTML, CSS, React, Socket Programming (WebSocket)
 - Grafana, Prometheus, AppSignal, New Relic
 - Learning: TypeScript, GraphQL, Jest, Kubernetes
- Flask, FastAPI, Shell Scripting, REST APIs
 - Redis, Sidekiq, Git, GitHub, OpenAI APIs
 - [Certified Amazon Web Services Solution Architect](#)
 - RSpec, PyTest, Microservices, Monorepo
 - Docker

Achievements

- Open Source contributor to [Polar.sh](#)
- Published manuscript on [Object Detection and Segmentation](#) in an International Scientific Research Journal.
- Successfully got the first startup acquired by Juspay

Education

Bachelor of Technology in Computer Science at Cambridge Institute of Technology (2016-2020) with GPA 3.9/5